YE CHENG

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Mail ⋄ Homepage ⋄ LinkedIn ⋄ GitHub

EDUCATION

Master of Informatics, Kyoto University, Graduate school of Informatics 2026.4 – Expected 2028.3

Bachelor of Engineering, Kansai University, Department of Civil, Environmental and Applied Systems Engineering 2021.4 – 2026.3

SKILL

Completed Course. Data Structure and Algorithms, Operating System, Introduction to Artificial Intelligence, Computer Network, Information Security, Information Theory, Database, and so on

Programming Skills. Python (Pandas, Matplotlib), Golang, C, Vimscript

Tools and Frameworks. Node.js, Git, Linux, PyTorch, Docker, Vim, PostgreSQL

Language Skills. English (TOEIC score: 880/990), Japanese (JLPT N1), Chinese (Native)

PROJECTS

Mogu Mogu. In this project, I was mainly responsible for building the backend and the overall system architecture of a meal and fitness tracking app using Go. The goal is to develop software that allows users to record daily activities and meals to receive personalized AI-generated nutrition recommendations. The main features include JWT token verification, user profile and preference management, meal and activity tracking, and dynamic prompt assembly for AI recommendations. Data persistence, session storage, and analytics are implemented using PostgreSQL, GORM, and strict middleware controls. The primary technology stack of this project includes Go, Gin, GORM, PostgreSQL, and JWT.

Simblock_go. In this project, I was responsible for designing and developing simblock_go, a Go-based high-performance replication of SimBlock, a blockchain network simulator originally developed by my affiliated laboratory (Shudo Laboratory). The goal is to migrate and optimize the platform to mimic block propagation and consensus dynamics with superior execution efficiency. The main features include a discrete-event simulation engine, network topology processing, Proof-of-Work (PoW) consensus, and an asynchronous message task dispatching system (handling Inv, Block, and CmpctBlock messages). Node routing and multi-region propagation are fully implemented through event timers and custom task queues. The primary technology stack includes Go, YAML configuration, and an integration testing framework.

Conference Paper. In this paper, I proposed and implemented GOFA, a novel gradient-oriented backdoor attack method designed to expose critical security vulnerabilities in Vertical Federated Learning (VFL) systems. The goal is to develop an offensive threat model that successfully injects backdoor triggers into localized inputs or intermediate embeddings without requiring any prior knowledge of task labels. The main features include a server-provided gradient feedback harvesting mechanism, automated poisoned dataset construction, and the integration of adversarial-example techniques (e.g., FGSM) to mask original features and strengthen trigger learning. The attack propagation, feature masking, and multi-party coordination are fully verified through specialized simulation pipelines across diverse datasets including CIFAR-10 and UCI-HAR. The primary technology stack of this project includes Python, PyTorch, TorchVision, and CUDA acceleration. Notably, the paper and findings of this research have been peer-reviewed and accepted by the International Conference on Electrical, Computer and Energy Technologies (ICECET).

EXTRA-CURRICULAR ACTIVITIES

- **2022–2023 Kura Sushi.**
  During my time at Kura Sushi, I worked as a part-time kitchen staff member where I was primarily responsible for preparing sushi and making tempura.